4. (i) null set (ii) null set (iii) singleton set (iv) null set

5. (i) overlapping (ii) disjoint (iii) overlapping

6. (i) {square, rhombus} (ii) {circle} (iii) {triangle} (iv) { }

7. { }, {a}, {a, b}, {a, {a, b}}

8. (i) {{ }, {a}, {b}, {a, b}}

(ii) {{ }, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3 }, {1, 2, 3}}

(iii) {{ }, {p}, {q} {r}, {s}, {p, q}, {p, r}, {p, s}, {q, r}, {q, s}, {r, s}, {p, q, r}, {p, q, s}, {p, r, s}, {q, r, s}, {p, q, r, s}} (iv) $P(E)$= {{ }}

9. (i) 8, 7 (ii) 1024, 1023

10. (i) 16 (ii) 1 (iii) 8

Exercise 1.3

1. (i) {2, 4, 7, 8, 10} (ii) {3, 4, 6, 7, 9, 11} (iii) {2, 3, 4, 6, 7, 8, 9, 10, 11}

(iv) {4, 7} (v) {2, 8, 10} (vi) {3, 6, 9, 11}

(vii) {1, 3, 6, 9, 11, 12} (viii) {1, 2, 8, 10, 12}

(ix) {1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12}

2. (i) {2, 5, 6, 10, 14, 16}, {2, 14}, {6, 10}, {5, 16}

(ii) {a, b, c, e, i, o, u}, {a, e, u}, {b, c}, {i, o}

(iii) {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, {1, 2, 3, 4, 5}, {6, 7, 8, 9, 10,}, {0}

(iv) {m, a, t, h, e, i, c, s, g, o, r, y}, {e, m, t,}, {a, h, i, c, s}, {g, o, r, y}

3. (i) {a, c, e, g} (ii) {b, c, f, g} (iii) {a, b, c, e, f, g} (iv) {c, g} (v) {c, g}

(vi) {a, b, c, e, f, g} (vii) {b, d, f, h} (viii) {a, d, e, h}

Answers_Combine.indd 306 02/12/2022 21:54:33